CS 431/631 Introduction to Big Data

Data Collection and Exploration

Some slides are borrowed from John Canny, Mohamed Eltabakh, et. al.

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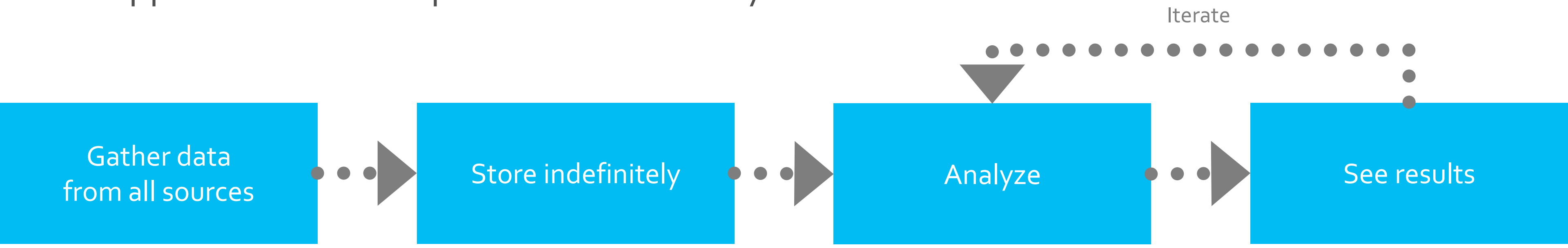
New big data thinking: All data has value

- All data has potential value
- Data hoarding
- No defined schema—stored in native format
- Schema is imposed and transformations are done at query time *(schema-on-read).*
- Apps and users interpret the data as they see fit



Big Data is driving transformative changes

	Traditional	Big Data
Data Characteristics	Relational (with highly modeled schema)	All Data (with schema agility)
Cost	Expensive (storage and compute capacity)	Commodity (storage and compute capacity)
Culture	Rear-view reporting (using relational algebra)	Intelligent action (using relational algebra AND ML, graph, streaming, image processing)

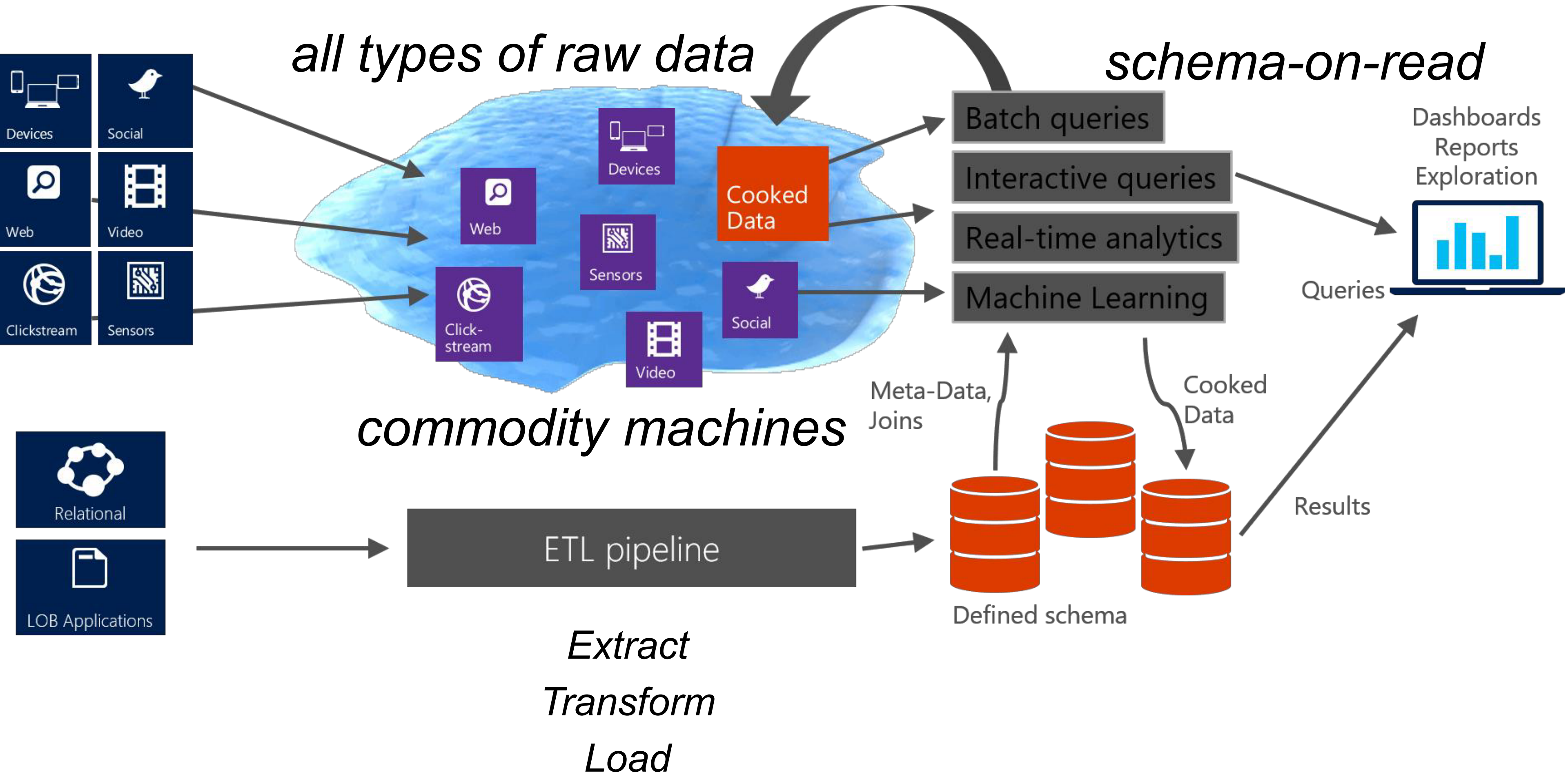
The (changing) role of Schema

Schema specify the **structure** and **types** of a data repository, e.g. the types of each column in a table.
They may also specify constraints **within** or **between** data fields.

Traditional databases are **schema-on-write**. You cannot load data into a table without a schema.

Newer data stores (e.g., NoSQL key-value store) are **schema-on-read** or **schemaless**: You can defer applying a schema until you read the data, or avoid schema altogether.
Usually, you need to follow certain formats (e.g., JSON, key-value, table with dynamic column, column family, version etc.)

The data lake and warehouse



Data Lake vs Data Warehouse

	Data Lake	Data Warehouse
1. Data Storage	A data lake contains all an organization's data in a raw, unstructured form, and can store the data indefinitely — for immediate or future use.	A data warehouse contains structured data that has been cleaned and processed, ready for strategic analysis based on predefined business needs.
2. Users	Data from a data lake — with its large volume of unstructured data — is typically used by data scientists and engineers who prefer to study data in its raw form to gain new, unique business insights.	Data from a data warehouse is typically accessed by managers and business-end users looking to gain insights from business KPIs, as the data has already been structured to provide answers to pre-determined questions for analysis.
3. Analysis	Predictive analytics, machine learning, data visualization, BI, big data analytics.	Data visualization, BI, data analytics.
4. Schema	Schema is defined after the data is stored in a data lake vs data warehouse, making the process of capturing and storing the data faster.	In a data warehouse, the schema is defined before the data is stored. This lengthens the time it takes to process the data, but once complete, the data is at the ready for consistent, confident use across the organization.
5. Processing	ELT (Extract, Load, Transform). In this process, the data is extracted from its source for storage in the data lake, and structured only when needed.	ETL (Extract, Transform, Load). In this process, data is extracted from its source(s), scrubbed, then structured so it's ready for business-end analysis.
6. Cost	Storage costs are fairly inexpensive in a data lake vs data warehouse. Data lakes are also less time-consuming to manage, which reduces operational costs.	Data warehouses cost more than data lakes, and also require more time to manage, resulting in additional operational costs.

What Makes a Successful Data Lake?

Right Platform + *Right Data* + *Right Interface*

Right Platform: Hadoop

- Volume - massively scalable
- Variety - schema on read
- Velocity - faster than conventional at scale
- Future Proof – modular – same data can be used by many different projects and technologies
- Platform cost – commodity machines, extremely attractive cost

https://www.youtube.com/watch?v=aReuLtY0YMI

Right Data Challenges: Most Data is Lost, so can't Analyze it Later

Data Exhaust

Only a Small Portion of Data is Saved in Enterprises Today in Data Warehouses

Right Data: Save <u>Raw</u> Data <u>Now</u> to Analyze Later

- Don't know now what data will be needed later
- Save as much data as possible <u>now</u> to analyze later
- Save <u>raw</u> data, so can be treated correctly for each use case

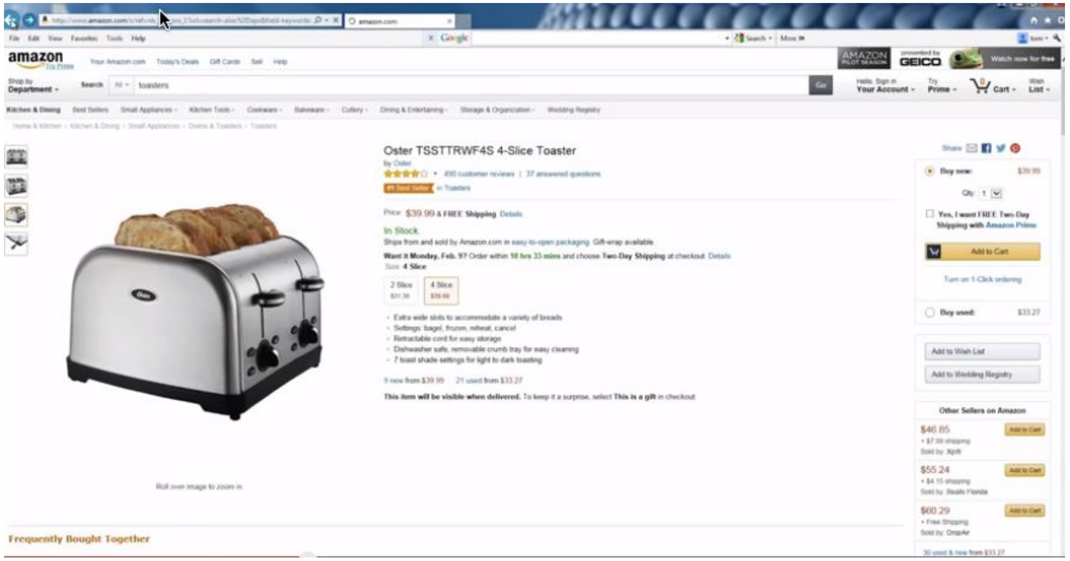
Right Data Challenges: Data silos and data hoarding

- Departments hoard and protect their data and do not share it with the rest of the enterprise

- Frictionless ingestion does not depend on data owners

Right Interface: Key to Broad Adoption

- Data marketplace for data self-service
- Providing data at the right level of expertise



Providing Data at the Right Level of Expertise

Clean, trusted, prepared data

Raw data

Data Scientists

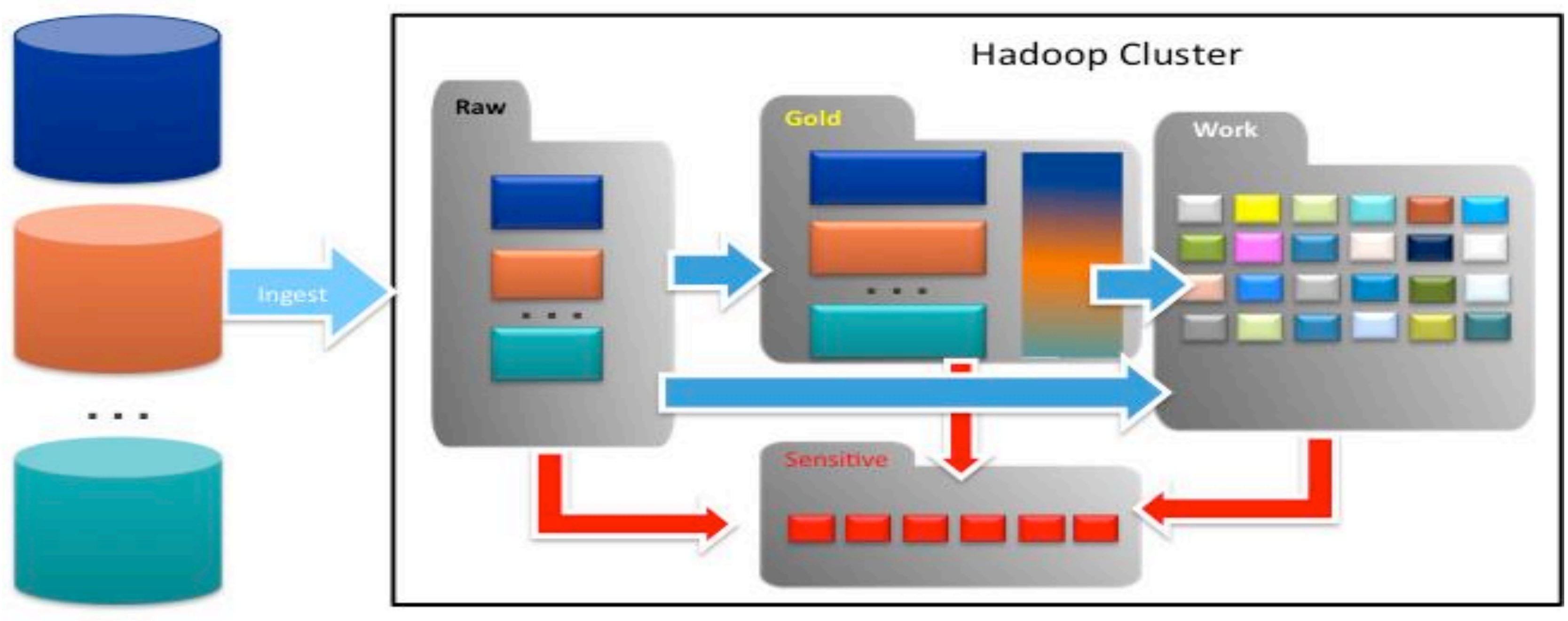
Business Analyst

Roadmap to Data Lake Success

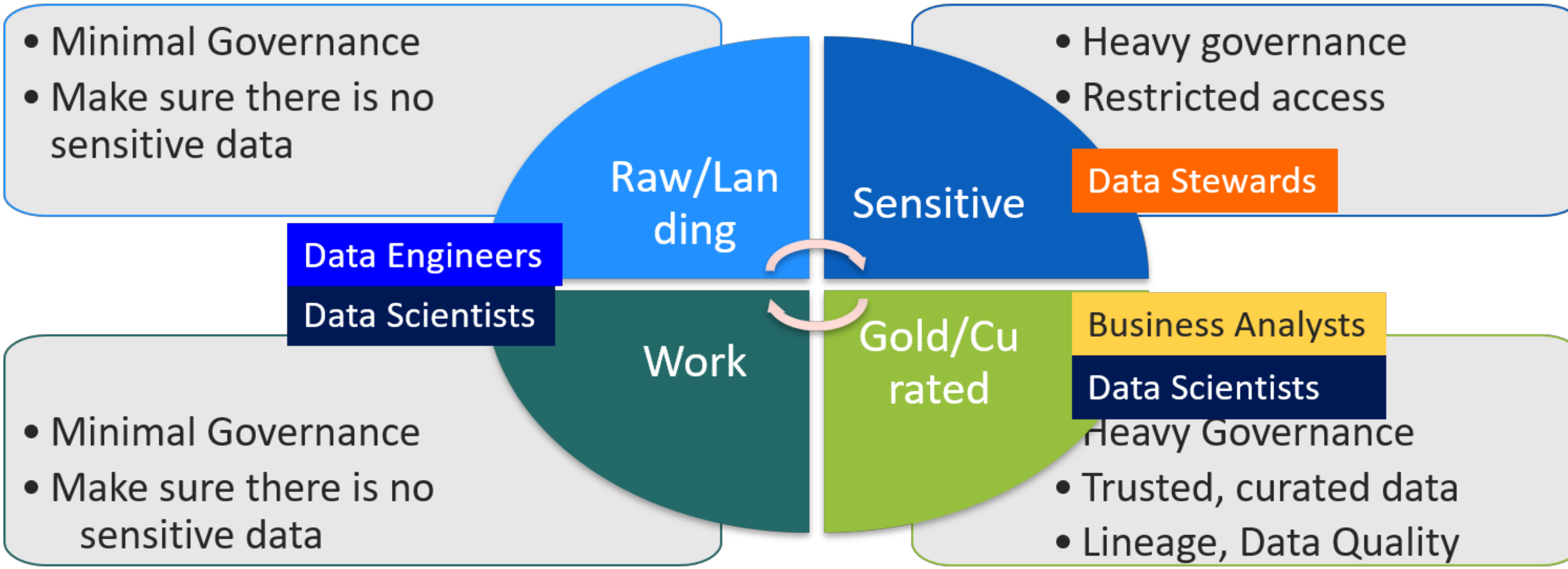
1. Organize the lake
2. Set up for Self-Service
3. Open the lake to the users

Organize Data Lake into Zones

Organize the lake



Multi-modal IT – Different Governance Levels for Different Zones



Data curation: user engaging, data integration, representation

Data lineage: data life cycle (origin, changes, errors)

Multi-modal IT with different governance levels in different zones

Successful Data Lake

Right Platform + *Right Data* + *Right Interface*

Data Collection and Exploration Workflow

Data Collection Exploration

Find and Understand → Provision → Prep → Analyze

Find and Understand

Finding and Understanding Data

1. New technology (e.g., devices, sensors)
2. Crowd-sourcing
3. Warehouse or Data Lake
4. Important Things
   - Curation
     - the process of creating, organizing and maintaining data sets
   - Lineage
     - the process of understanding, recording, and visualizing data
   - Data quality
   - Governance

Data Sources at Web Companies

Find and Understand

- Examples from Facebook
  - Application databases
  - Web server logs
  - Event logs
  - API server logs
  - Ad server logs
  - Search server logs
  - Advertisement landing page content
  - Graph data
  - Images and video

Structured Data

Semi-structured Data

Unstructured Data

Provision

Accessing and Provisioning Data

Cannot give all access to all users
Must protect PII data and Sensitive information
(PII: personally identifiable information)

Top down approach

- Find and de-identify all sensitive data
- Provide access to every user for every dataset as needed

fast, one-time effort

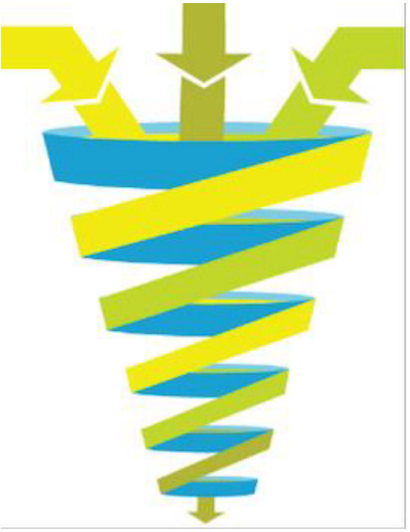
Agile/Self-Service Approach

- Create a metadata only catalog
- When users request access, data is de-identified and provisioned

scalable, dynamic data

Prep

Data Preparation



- Prepare data for analytics
  - Clean data
    - Remove or fix bad data
    - Fill in missing values
    - Convert to common units of measure
  - Shape data
    - Combine (join, merge)
    - Transform (aggregate, filter, bucketize etc.)
    - Normalization/standardization
  - Blend data - harmonize data from multiple sources to a common schema/model

- Tooling
  - Data cleaning tools
  - Visualization tools
  - SQL languages for structured data
  - Hadoop and scripting for unstructured data

Prep

Data Preparation

- ETL
  - We need to **extract** data from the **source(s)**
  - We need to **load** data into the **sink**
  - We need to **transform** data at the source, sink, or in a **staging area (intermediate)**

  - Sources: file, database, event log, web site, HDFS…
  - Sinks: Python, R, SQLite, RDBMS, NoSQL store, files, HDFS…

Analyze

Data Analysis

- Many wonderful Visualization tools
- Statistical tools
- Data mining and machine learning tools

Descriptive: e.g., median; describes data you have but can't be generalized beyond that: Statistics

Inferential: e.g., regression; enables inferences about the population beyond our data: Machine Learning and Prediction

Dirty Data Problems

1) Parsing text into fields (separator issues)
2) Naming conventions: ER: NYC vs New York
3) Missing required field (e.g., key field)
4) Different representations (2 vs Two)
5) Fields too long (get truncated)
6) Primary key violation (from un- to structured or during integration)
7) Redundant Records (exact match or other)
8) Formatting issues – especially dates
9) Licensing issues/Privacy/ keep you from using the data as you would like

Dirty Data

- The Statistics View:
  - There is a process that produces data
  - We want to model ideal samples of that process, but in practice we have non-ideal samples:
    - **Distortion** – some samples are corrupted by a process
    - **Selection Bias** - likelihood of a sample depends on its value
    - **Left and right censorship** – partially known
    - **Dependence** – samples are supposed to be independent, but are not (e.g. social networks)
  - You can add new models for each type of imperfection, but you can't model everything.
  - What's the best trade-off between accuracy, performance, simplicity?

Dirty Data

- The Database View:
  - I got my hands on this data set
  - Some of the values are missing, corrupted, wrong, duplicated
  - You get a better answer by improving the quality of the values in your dataset

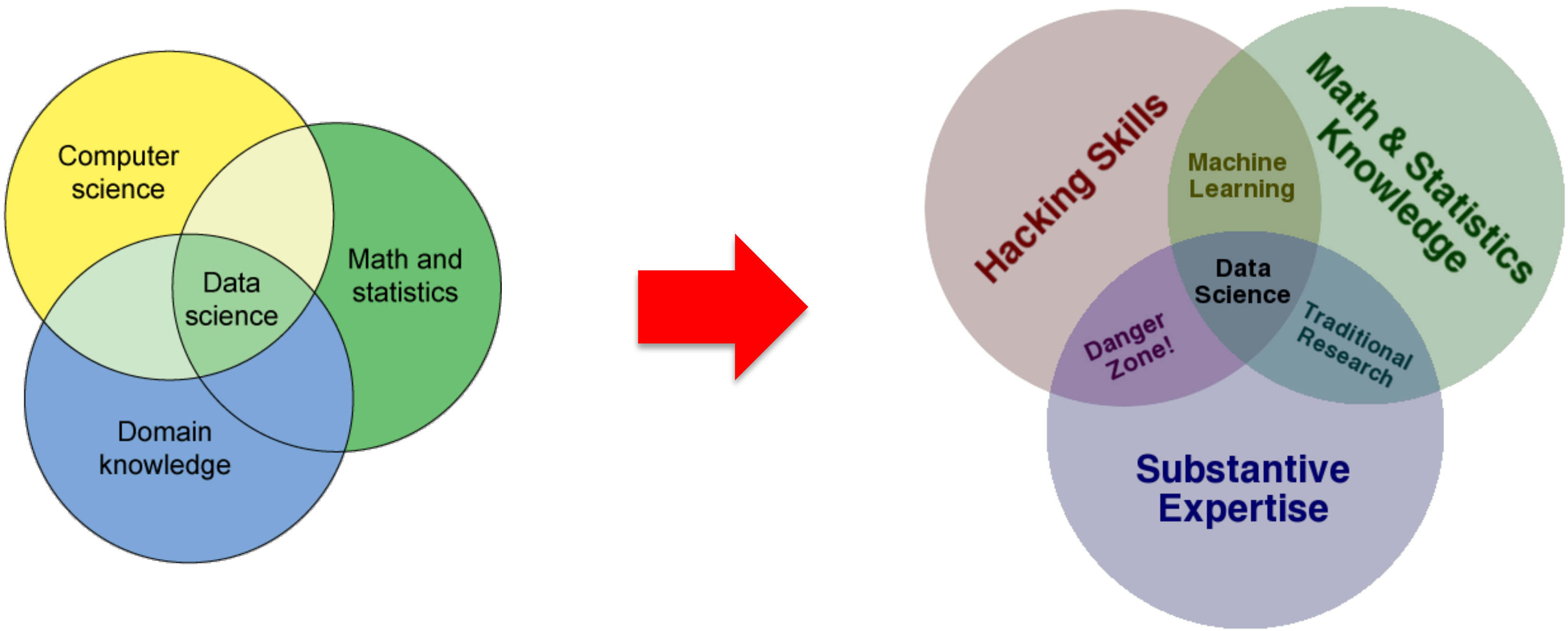
Dirty Data

- The Domain Expert's View:
  - This Data Doesn't look right
  - This Answer Doesn't look right
  - What happened?
- Domain experts have an implicit model of the data that they can test against…

Dirty Data

- The Data Scientist's View:

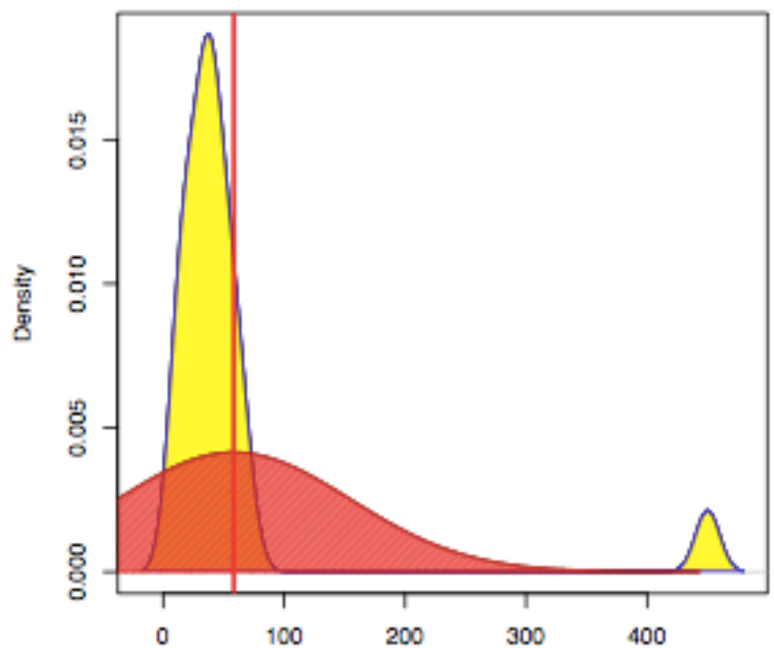
  Some Combination of all of the above



Danger of Numeric Outliers

12	13	14	21	22	26	33	35	36	37	39	42	45	47	54	57	61	68	450

ages of employees (US)



- median 37
- mean 58.52632
- variance 9252.041

Challenges with Sensor Data

He walks through walls;

He flies across the room…

Original Data

Too much cleaning and you lose detail.

Ubisense tracking data from Ryan Appierspach

Where can Dirty Data Arise?

- (Source) Data is dirty on its own.
- Transformations corrupt the data (complexity of software pipelines).
- Data sets are clean but integration (i.e., combining them) screws them up.
- "Rare" errors can become frequent after transformation or integration.
- Data sets are clean but suffer "bit rot"
  - Old data loses its value/accuracy over time
- Any combination of the above

Conventional Definition of Data Quality

- Accuracy
  - The data was recorded correctly.
- Completeness
  - All relevant data was recorded.
- Uniqueness
  - Entities are recorded once.
- Timeliness
  - The data is kept up to date.
    - Special problems in federated data: time consistency.
- Consistency
  - The data agrees with itself.

Problems …

- Unmeasurable
  - Accuracy and completeness are extremely difficult, perhaps impossible to measure.
- Context independent
  - No accounting for what is important. E.g., if you are computing aggregates, you can tolerate a lot of inaccuracy.
- Incomplete
  - What about interpretability, accessibility, metadata, analysis, etc.
- Vague
  - The conventional definitions provide no guidance towards practical improvements of the data.

Finding a modern definition

- We need a definition of data quality which
  - Reflects the **use** of the data
  - Is **measurable** (we can define metrics)
    Leads to **improvements in processes**

- First, we need a better understanding of how and where data quality problems occur
  - The data quality continuum

Meaning of Data Quality

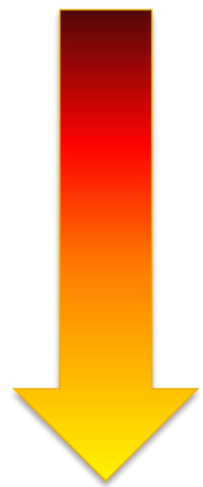
- There are many types of data, which have different uses and typical quality problems
  - Federated data
  - High dimensional data
  - Descriptive data
  - Longitudinal data
  - Streaming data
  - Web (scraped) data
  - Numeric vs. categorical vs. text data vs. image/voice/video

Meaning of Data Quality

- There are many uses of data
  - Operations (different importance, dependency)
  - Aggregate analysis (higher chance of glitched data, sometimes better tolerance)
  - Customer relations (accurate fields) …
- Data Interpretation: the *rules* behind the data.
- Data Suitability: can you get the answer from the available data?
  - Use of proxy data
  - Relevant data is missing
  - Fairness

The Data Quality Continuum

- Data and information is not static, it flows in a data collection and usage process
  - Data gathering
  - Data delivery
  - Data storage
  - Data integration
  - Data retrieval
  - Data mining/analysis



Data Gathering

- How does the data enter the system?
- Sources of problems:
  - Manual entry
  - No uniform standards for content and formats
  - Parallel data entry (duplicates)
  - Approximations, surrogates – SW/HW constraints
  - Measurement or sensor errors.

Data Gathering - Solutions

- Potential Solutions:
  - Preemptive:
    - Process architecture (build in integrity checks)
    - Process management (reward accurate data entry, data sharing, data stewards)
  - Retrospective:
    - Cleaning focus (duplicate removal, merge/purge, name & address matching, field value standardization)
    - Diagnostic focus (automated detection of glitches).

Data Delivery

- Destroying or mutilating information by inappropriate pre-processing
  - Inappropriate aggregation
  - Nulls converted to default values
- Loss of data:
  - Buffer overflows
  - Transmission problems
  - No checks

Data Delivery - Solutions

- Build reliable transmission protocols
  - Use a relay server
- Verification
  - Checksums, verification parser
  - Do the uploaded files fit an expected pattern?
- Relationships
  - Are there dependencies between data streams and processing steps
- Interface agreements
  - Data quality commitment from the data stream supplier.

Data Storage

- You get a data set. What do you do with it?
- Problems in physical storage
  - Can be an issue, but terabytes are cheap.
- Problems in logical storage
  - Poor metadata.
    - Data feeds are often derived from application programs or legacy data sources. What does it mean?
  - Inappropriate data models.
    - Missing timestamps, incorrect normalization, etc.
  - Ad-hoc modifications.
    - Structure the data to fit the GUI.
  - Hardware / software constraints.
    - Data transmission via Excel spreadsheets, Y2K, exceed data type limit, round off.

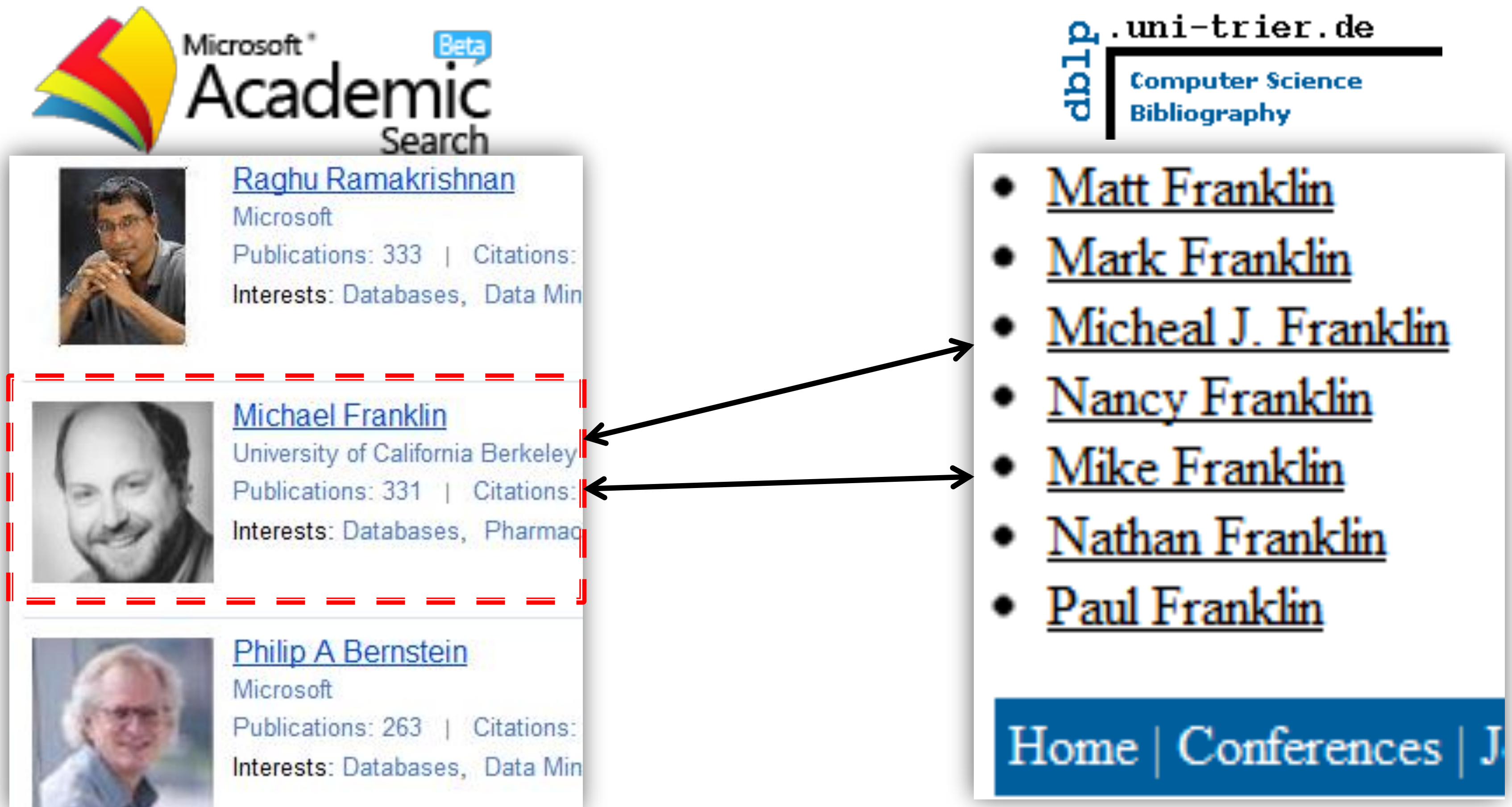
Data Storage - Solutions

- Metadata
  - Document and publish data specifications.
- Planning
  - Assume that everything bad will happen.
  - Can be very difficult.
- Data exploration
  - Use data browsing and data mining tools to examine the data.
    - Does it meet the specifications you assumed?
    - Has something changed?

Data Integration

- Combine data sets (acquisitions, across departments).
- Common source of problems
  - Heterogenous data : no common key, different field formats
    - Approximate matching
  - Different definitions
    - What is a customer: an account, an individual, a family, …
  - Time synchronization
    - Does the data relate to the same time periods? Are the time windows compatible?
  - Legacy data
    - Spreadsheets, ad-hoc structures
  - Sociological factors
    - Reluctance to share

Example: Data Integration



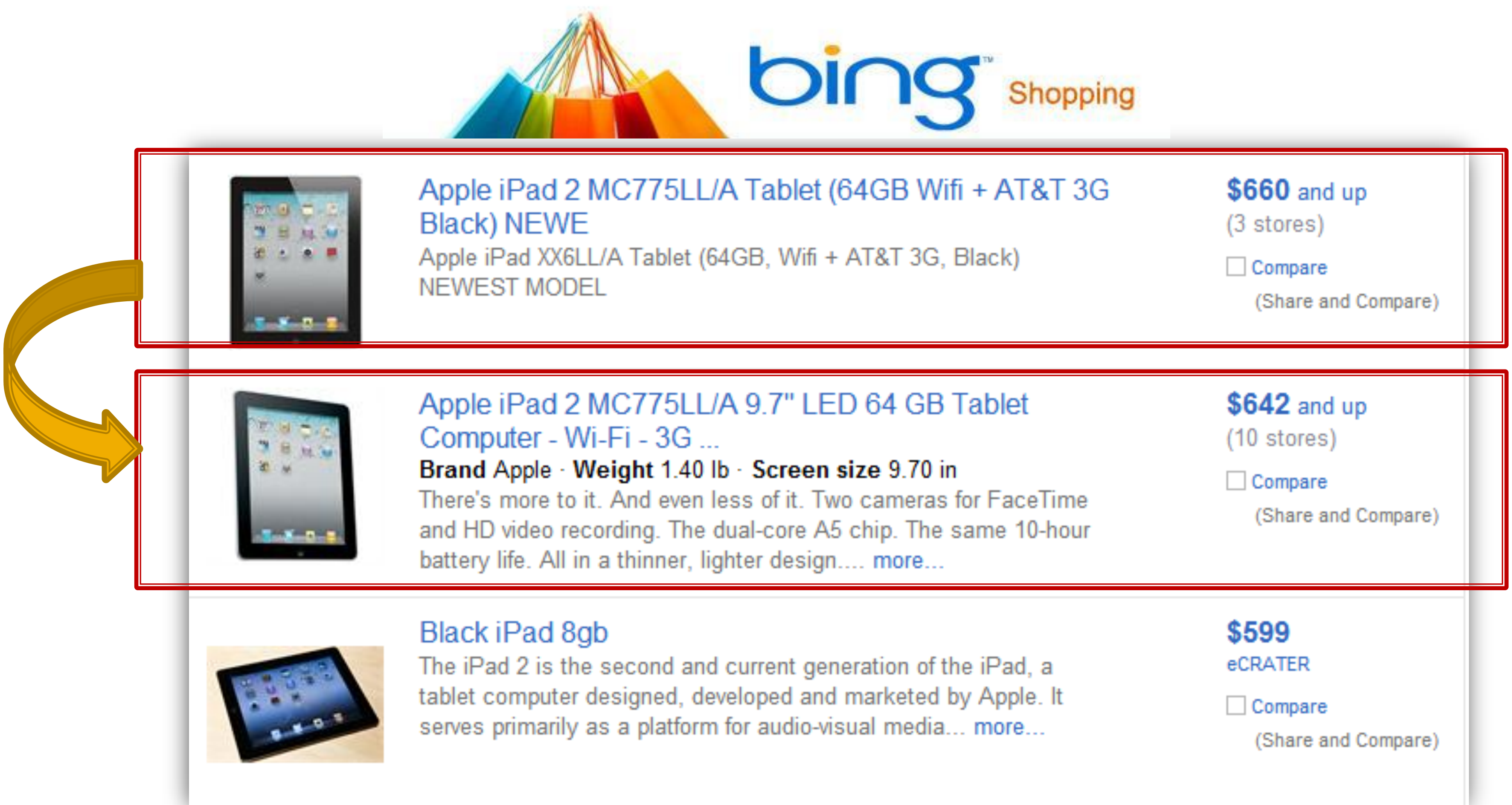
Approximate Matching

- Relate tuples whose fields are "close"
  - Approximate string matching
    - Generally, based on edit distance (Julien -> Julian, change e -> a, so distance = 1)
    - *q-gram* (n-gram): e.g., AGCTTCGA: 1-gram: A, G, C, T, T, C, G, A; 2-gram: AG, GC, CT, TT, TC, CG, GA; 3-gram: AGC, GCT, CTT, TTC, TCG, CGA
  - Approximate tree matching
    - For Nested Data Structures (or flattened ones)
    - Much more expensive than string matching
    - Recent research in fast approximations
  - Feature vector matching
    - Similarity search
    - Image classification
  - Ad-hoc or Domain-focused matching
    - Use domain insights and/or clever tricks

Duplicate Record Detection needs DeDup!

- Resolve multiple different mentions:
  - Entity Resolution
  - Reference Reconciliation
  - Object Identification/Consolidation
- Remove Duplicates
  - Merge/Purge
- Record Linking (across data sources)
- Approximate Match (accept fuzziness)
- …

Example: DeDup/Cleaning



Data Integration - Solutions

- Commercial Tools (IBM InfoSphere, Xplenty, SSIS)
  - Significant body of research in data integration
  - Many tools for address matching, schema mapping are available.
- Data browsing and exploration
  - Many hidden problems and meanings : must extract metadata.
  - View before and after results : did the integration go the way you thought?

Data Retrieval

- Exported data sets are often a view of the actual data. Problems occur because:
  - Source data not properly understood.
  - Need for derived data not understood.
  - Just plain mistakes.
    - Inner join vs. outer join (inner join + rows w/o match)
    - Understanding NULL values
- Computational constraints
  - E.g., too expensive to give a full history, we'll supply a snapshot.
- Incompatibility
  - Ebcdic? Unicode?

Data Mining and Analysis

- What are you doing with all this data anyway?
- Problems in the analysis.
  - Scale and performance
  - Confidence bounds?
  - Black boxes and dart boards
  - Assumptions
  - Insufficient domain expertise
  - Casual empiricism

Retrieval and Mining - Solutions

- Data exploration
  - Determine which models and techniques are appropriate, find data bugs, develop domain expertise.
- Ablation study
  - Remove components and study the performance.
- Continuous analysis
  - Are the results stable? How do they change?
- Accountability
  - Make the analysis part of the feedback loop.

Data Quality Constraints

- Many data quality problems can be captured by *static* constraints based on the schema.
  - Nulls not allowed, field domains, foreign key constraints, etc.
- Many others are due to problems in workflow, and can be captured by *dynamic* constraints
  - E.g., orders above $200 are processed by Biller 2
- The constraints follow an 80-20 rule
  - A few constraints capture most cases, thousands of constraints to capture the last few cases.
- Constraints are measurable. Data Quality Metrics?

Data Quality Metrics

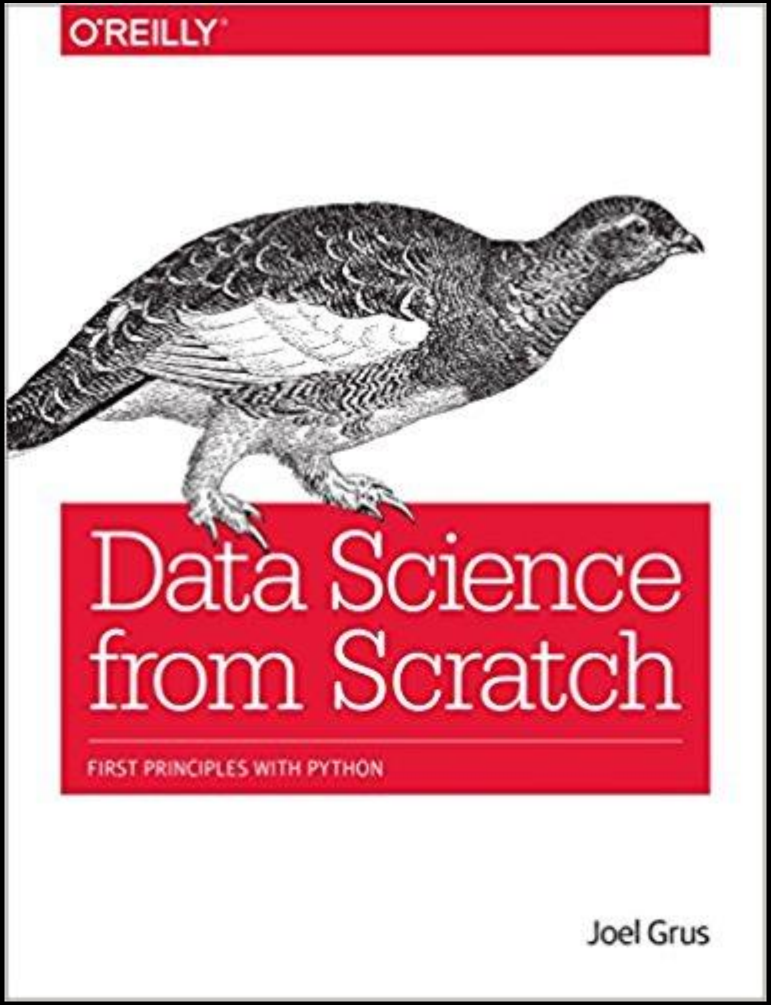
- We want a measurable quantity
  - Indicates what is wrong and how to improve
  - Realize that DQ is a messy problem, no set of numbers will be perfect
- Types of metrics
  - Static vs. dynamic constraints
  - Operational vs. diagnostic
- Metrics should be *directionally correct* with an improvement in use of the data.
- A very large number metrics are possible
  - Choose the most important ones.

Examples of Data Quality Metrics

- Conformance to schema
  - Evaluate constraints on a snapshot.
- Conformance to rules
  - Evaluate constraints on changes in the database.
- Accuracy
  - Perform inventory (expensive), or use proxy (track complaints). Audit samples?
- Accessibility
- Interpretability
- Glitches in analysis
- Successful completion of end-to-end process

AWS re:Invent 2020: An introduction to data lakes and analytics on AWS

https://www.youtube.com/watch?v=V2tV4aa_x8U



Extended Reading